RADIANCEENGINE · GAME DESIGN ROADMAP

Spellbrawl: design direction and roadmap

18 October 2026 · Follows "Spellbrawl: what to change and improve" (13 October) · Based on the first round of playtest feedback

Spellbrawl is a party brawler for 2 to 12 players. Each mage brings three spells from a pool of nine, plus the shared Ember Bolt, and tries to shove the others off a crumbling basalt island into lava. Spells don't hurt. Lava does, and the lower your health, the further you fly. Playtesters say the game looks a bit dead, spell selection is confusing and boons go unnoticed. They also want more variety, randomness and long-term goals. This document sets the direction for all of that and the order to build it in.

One idea runs through everything below: every round ends with one big, visible, exciting choice, and everything on screen reads at a glance.

1 The plan at a glance

1. Replace the between-round shop with a single card pick, and edit spells by dragging them onto the hotbar. §4, §5
2. Colour pass: player colours, a cool counter-colour on the Caldera, calmer lava, a brighter hub. §3
3. Server-side bots, then overnight balance runs. Carried over from the 13 October roadmap. §6
4. First wave of new spells (Chain Hook, Firebomb, Splitting Bolt, Storm Strike) and upgrade cards for the ten existing spells. §7
5. Frozen Lake, the first map with its own mechanic, and a random map per match. §8
6. Steam: lobbies, invites, relay servers, rich presence and achievements. §11
7. Cosmetics and sideways unlocks. §10
8. Quick modes (Teams, Cursed Bomb, Spellball), then the co-op roguelike. §9

2 Feedback and where things stand

What players said	What is behind it	Section
"It looks a bit dead. Party games are more colourful."	The Caldera uses a single hue: 97% of its coloured pixels are orange. The hub is dark and grey: half its pixels have almost no colour. On both maps players are dark robes on dark ground.	§3
"Spell selection is confusing."	Changing a spell is "click a slot, then a spell", in a window shared with the boons.	§5
"I have not once selected a boon."	Boons sit in a side column next to the spell grid, in the same 12 seconds. They are small stat bumps (+7% move speed), and the round winner never gets one.	§4
"The spells are fine, just a bit unbalanced."	Nothing is measured yet: no bots, no per-spell statistics.	§6
"Add randomness, progression, cosmetics, achievements, new modes and more varied spells."	Covered by the card pick, mutators, map rotation, sideways unlocks, the Steam work and the new spell list.	§6–§11

Done since 13 October DONE

- The Caldera island is built from Voronoi cells in faceted chunks that tilt and sink.
- Emberhold is a floating-island lobby with practice dummies, the Spellbook, the Caldera Gate and secrets.
- Your spell loadout is saved between matches.
- The player hero has a realistic motion-capture animation set: idles, walk, jog and run, nine casts, dodges and hits.
- A Wardrobe offers three headwear options: hood, pointed hat or bare head.

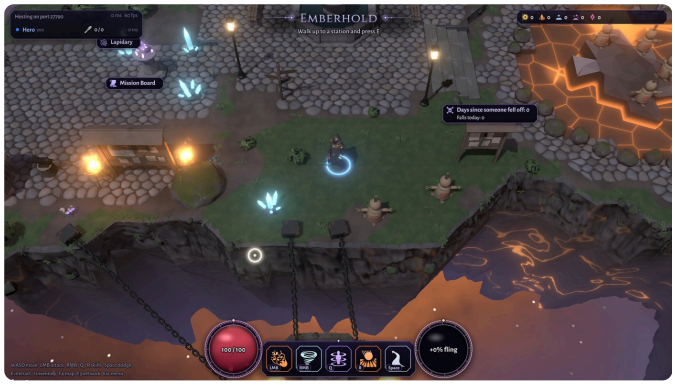
Still open from 13 October

- Bots, which are still the biggest risk-reducer.
- Momentum transfer between flung players, and deflecting projectiles.
- Spell augments. They become the card pick in §4.
- New spells, Rising Tide and team modes.
- Steamworks networking and the store page.

3 Colour: from “dead” to vibrant



Caldera: 98% of pixels are coloured; 97% of those lie between 0° and 45° (orange). Mean brightness 0.55.



Hub: only 48% of pixels are coloured. Mean saturation 0.22, mean brightness 0.32.

Figure 1. The game today, Release build at 1600 × 900. Left: a four-player bot match on the Caldera. Right: the Emberhold hub. The bars show how the coloured pixels (saturation of at least 0.2 and brightness of at least 0.12, with the HUD cropped out) spread around the colour wheel in 15° steps. The bars use the same scale in both charts.

What the numbers say

- **The Caldera isn't grey, it's monochrome.** Lava, rock, light and even shadows are all orange, because the lava is also the main light source. Nothing cool on screen makes the warm colours pop.
- **The hub has the opposite problem.** Dusk light makes it dark and grey, and the colour it does have (warm lamps, green grass, violet sky) is muted.
- **Players don't stand out on either map.** They are dark robes on dark ground, and their colour appears only on the name, the score dot and the ground ring.

What to change

1. **Player colours.** Give each player one strong colour, from the palette below. Use it on the robe trim, the hood or cape lining, the ground ring, the name and the core of their spells, plus a soft rim light. This adds the most colour for the least work and answers “where am I?” at once. Keep away from lava orange, and pair each colour with a ring shape so colour-blind players can tell everyone apart.
2. **A counter-colour on every map.** On the Caldera:
   - the island top becomes cool blue-grey obsidian with teal or violet crystal veins;
   - a blue sky fill turns shadows cool instead of orange;
   - the lava stays in bright cracks and rivers instead of large blown-out sheets.

3. **A brighter hub.** Raise the sun and sky light, saturate foliage, banners and flowers, and lift the shadows. Keep the arcane dusk in the sky, not on the ground.
4. **Colour grading, not tonemapping.** Add mid-tone saturation in the existing colour grade, with slightly cool shadows and warm highlights. Keep the soft-clip tonemapper, because filmic curves washed out the spell effects.
5. **Spell colours by element:** fire orange, frost cyan, lightning yellow-white, wind pale green, gravity indigo and arcane violet. Check every spell against every map: orange fire on orange lava disappears today.

Player palette: Azure Rose Lime Gold Violet Teal Magenta Silver (eight for now; extend to twelve for full lobbies)

Done when

Measure with the same histogram as Figure 1.

- **Caldera:** the largest 60° hue band holds at most 75% of the coloured pixels, and a second hue family holds at least 15%.
- **Hub:** at least 65% of pixels are coloured, and mean brightness is at least 0.40.
- **Players:** someone who has never played can point at every player in a fight screenshot within a second.

4 Between rounds: one card pick replaces the shop

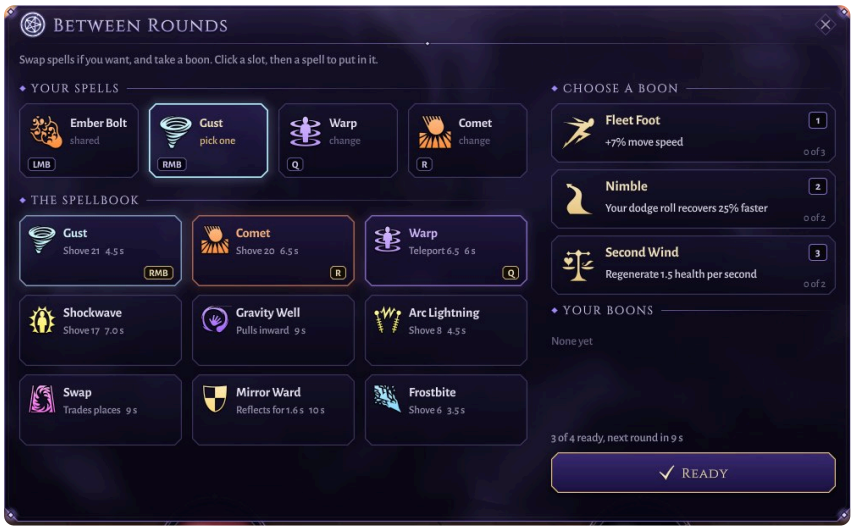


Figure 2. The between-round window today. Spells take the whole left side and the boons sit in a narrow column on the right, in the same 12 seconds. The round winner gets no boon at all.

Why nobody picks boons. Boons compete with spell swapping for the same 12 seconds, and spells win. They are small numbers (+7% move speed, shoves 12% harder) that you can't see in a fight. And the round winner, often the most engaged player, never gets one.

The new flow

Round result · 3.5 s → Card pick · 10 s → Countdown · 3 s 16.5 s between rounds, down from 18.5 s today

- **Who picks:** every player except the round winner picks one of three cards. The cards are large and centred. Spell editing sits in a strip below them (§5), so the two never compete for attention.
- **Catch-up:** the further behind on points a player is, the better their odds of rare cards.
- **Offers:** the server deals them, seeded per round. Upgrades are only offered for spells you own.
- **Timeout:** if the clock runs out, the left card is taken.
- **Visibility:**
  - Everyone sees everyone's picks in a feed, for example “Quill took Splitting Ember”.
  - Cards a player owns show as small icons under their name and on the scoreboard.

- **The winner's moment:** a crown and a short victory pose while the others pick.

Card type	What it does	Share of offers
Upgrade	Changes how one of your spells works. Each spell can take up to two.	about 70%
New spell	Offers a spell you don't have yet. Taking it swaps it into a slot.	about 20%
Wild	Bends a rule, often at a cost. Rare.	about 10%

Example upgrades

Spell	Upgrade cards
Ember Bolt	**Splitting Ember:** splits into three at the end of its range.
Gust	**Lingering Gale:** leaves a whirlwind that keeps pushing for 1.5 s. **Deflecting Gale:** bats projectiles back at whoever cast them.
Comet	**Shattering Impact:** the crater breaks the cells it lands on. **Twin Comets:** a smaller second comet lands 0.5 s later.
Warp	**Afterimage:** leaves a shockwave where you left.
Shockwave	**Aftershock:** a second, weaker pulse follows 0.6 s later.
Gravity Well	**Implosion:** when the well closes, it flings everyone outward.
Arc Lightning	**Pinball:** each jump knocks its target away from the previous one. **Overcharge:** one more jump.
Swap	**Heart Swap:** also swaps health, and with it the fling scaling.
Mirror Ward	**Thorned Mirror:** reflected spells come back 50% stronger.
Frostbite	**Black Ice:** a chilled target slides 40% further.

Wild card	Effect	Wild card	Effect
Glass Cannon	Your spells shove 40% harder, but you fly 30% further.	**Featherfall**	Once per round, your first touch of lava bounces you back to the nearest stone.
Echo	The first spell after a dodge casts twice at half strength.	**Heavy Robes**	You are shoved 25% less, but move 15% slower.

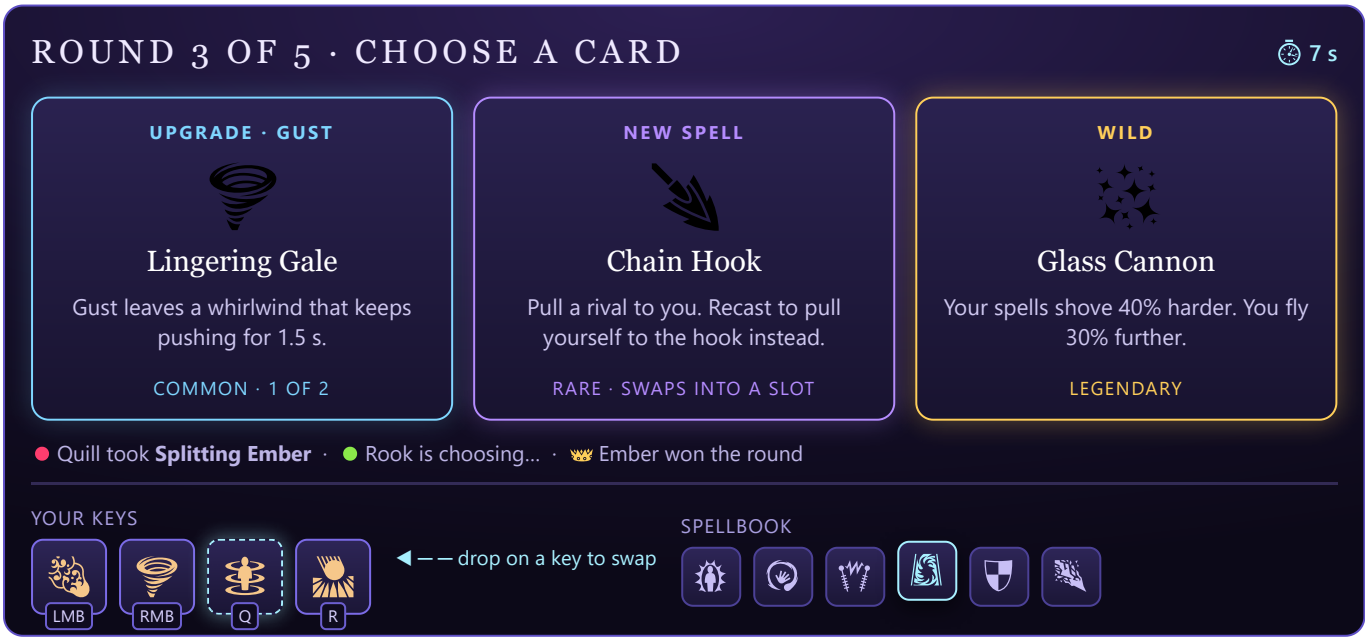


Figure 3. Proposed between-round screen. The card pick is the centrepiece, and the feed shows what everyone else takes. Spells are edited below it: drag one from the Spellbook strip onto a key, here Swap onto Q. Clicking a spell and then a key does the same, which also works on a controller. Rarity shows as the frame colour: common is cyan, rare violet, legendary gold.

5 Loadout: drag spells onto the hotbar

Today a spell is changed by clicking a slot and then a spell, inside a window, and the first round is preceded by a 20-second draft.

1. **The hotbar is the drop target.** Drag a spell from the Spellbook onto RMB, Q or R. Dropping on a filled key swaps the two spells, and dragging between keys reorders them.
2. **Click a spell, then a key** as the fallback. It also works on a controller and the Steam Deck.
3. **Every spell card** shows its icon, name, one action word (Push, Pull, Blink, Swap, Block, Zone, Slow), one line of text and its cooldown. Later, a two-second looping preview on hover.
4. **Matches start with the loadout saved in the Emberhold Spellbook.** The 20-second draft becomes a ready check, with "Change spells" as an option rather than a requirement.
5. **Between rounds, the Spellbook strip sits under the cards** (Figure 3), so picking a card and editing spells never fight for the same space.

6 Randomness, balance and bots

Put randomness in the choices and the arena, never in the outcome. Card offers, mutators and the map are random. Whether a hit lands and how far it sends someone are not, so every ring-out stays readable and earned.

Mutator	What changes
Ice Floor	Much less friction, so every shove slides about 1.6 times as far.
Hurricane	Every push is 1.5 times as strong.
Meteor Shower	Comets with clear warnings land on random cells.
Darkness	The lava is the only light, which shows off the radiance cascades.
Mirror Match	Everyone gets the same random loadout each round.
Quack Hour	Every Ember Bolt hit turns its target into a duck for half a second.

An event round, with a random mutator every third round, makes them part of a normal match instead of a separate menu.

Balance: measure first

- **Measure:** run bot-against-bot matches overnight. For each spell, report:
  - pick rate;
  - ring-outs per minute;
  - relative win rate, which is the win rate of players who own the spell divided by the average.

  Keep every spell between 0.9 and 1.1.
- **Principles:**
  - each spell does one thing;
  - instant displacement gets a visible warning;
  - stronger spells get longer cooldowns;
  - every spell has a written counter.
- **Watch list,** to measure rather than weaken blindly:
  - Swap and Warp, which give no warning;
  - Gravity Well followed by Comet;
  - Mirror Ward against bolt-heavy loadouts.
- **Bots** (§7 of the 13 October roadmap): a server-side brain writes ordinary player input. Knockback distance has a closed form, $d = v_0 / 3.2$, so the bot can score casts and positions directly. Bots fill empty lobbies and run the balance matches.

7 Spells: new verbs

Today's ten spells mostly push. That is the core of the game, but it leaves gaps: pulling one rival, terrain, rescue, sustain, tricky paths (lobbed, split, bouncing) and anything funny.

Spell today	Action	Notes
Ember Bolt	Push, projectile	Shared basic attack on LMB, push 11.5, 0.9 s cooldown.
Gust	Push, cone	The strongest single shove (21), at close range.
Comet	Push, delayed area	Lands 0.85 s after casting, radius 2.4.
Shockwave	Push, around you	Radius 3.3, 7 s cooldown.
Arc Lightning	Push, chain	Jumps between up to three rivals, weak per hit (8).
Frostbite	Slow	A small push (6) and a slow.
Gravity Well	Pull, area	Drags everyone inside to its centre.
Warp	Move yourself	The way back out of the lava.
Swap	Move a rival	Instant displacement with no warning.
Mirror Ward	Block and reflect	1.6 s of immunity to shoves, and spells bounce back.

New spell	Action	How it works	Counter	Size
Chain Hook WAVE 1	Pull one rival, or move yourself	Throw a hook that pulls a rival to you. Recast within a second to yank yourself to the hook instead. Pull, dodge, and they fly past you into the lava.	The hook is slow: sidestep it. Mirror Ward sends it back.	M
Firebomb WAVE 1	Lobbed area	Arcs to the cursor, skips once, and pushes on both landings.	Both landing circles show in advance.	S
Splitting Bolt WAVE 1	Push, trick	A slow bolt. Recast to split it into three spreading shards.	Watch the slow bolt. Each shard is weak.	M
Storm Strike WAVE 1	Fast, small area	Lightning hits the cursor 0.3 s after casting, then jumps once to the nearest rival. Comet is the opposite: slow (0.85 s) and big.	Keep moving, and don't bunch up.	S
Basalt Pillar	Terrain	Raises a pillar for 4 s that blocks shots and stops knockback. Bolts can bounce off it.	Go around it. Comet breaks it.	L
Ricochet	Bounce	Bounces off pillars, cliffs and players up to three times. Best on maps with obstacles.	The angles are readable. Dodge the last bounce.	M
Leech	Drain	A beam you hold for up to 1.5 s. It steals health, so you fly less and they fly more.	Walk out of range to break the beam.	M
Recall	Rescue	Snaps you back to where you were 1.5 s ago.	Punish the spot they will return to.	M
Quack	Transform	Turns a rival into a duck for 1.2 s: no casting, and they slide 30% further. The party spell, and the duck model already exists.	Mirror Ward, or dodge the slow projectile.	M

Rules for every new spell: it adds a new action or a new way to use one, it has a visible warning, and it has a counter.

Backlog from 13 October: Fissure, Obsidian, Boomerang, Homing Wisp, Geyser and Rush.

Sizes are rough, for one developer: S a few days, M one to two weeks, L a month or more.

8 Maps: one mechanic and one colour each

Map	Colours	Mechanic	Builds on	Size
Caldera (today)		The crumbling island. Gets the cool counter-colour from §3.	Colour pass only.	S
Frozen Lake		Ice: shoves slide much further. Cracked ice gives way into freezing water, which takes the role of lava.	Per-cell friction and the collapse plan.	M
Sky Ruins		No lava: you fall. Platforms drift, and wind gusts, announced by streaks, push everyone one way.	Emberhold's floating-island tech and void falls.	L
Crystal Cavern		Crystal pillars reflect spells, and the crystals light the cave.	Mirror Ward's reflect code and the radiance cascades.	M
Storm Peak		Lightning strikes cells after a warning, and the safe zone moves.	Comet's ground strikes and the collapse warnings.	M
Rising Tide		Terraces. The lava rises in steps.	§6 of the 13 October roadmap.	L

- **Map rotation:** a random map per match adds randomness at no cost. Later, a "Tour" playlist can change the map every round.
- **Keep world state a function of seed and clock,** as the collapse is today. Only terrain changes made by players need replicating.

9 Modes and the co-op roguelike

Mode	Twist	Size
Teams 2v2 / 3v3	Pull allies out of the lava with Chain Hook or Gravity Well. Team scores. Probably the competitive format, while free-for-all stays the party default.	M
Cursed Bomb	A bomb sticks to one player and jumps to whoever they hit. It explodes after 10 s with a huge fling.	S
Spellball	Knock a heavy ball into the other team's lava pit.	M
King of the Crater	Score while standing in the shrinking centre.	S
Mirror Match	Everyone gets the same random loadout each round. Also works as a mutator.	S

Caldera Trials: the co-op roguelike

Arena 1 → Card → Arena 2 → Card → Elite arena → Card → … → Boss → Unlocks

- **Players and goal:** 1 to 4 players. Monster waves climb onto the island, and you knock them into the lava.
- **Reused systems:**
  - The same card pick as PvP runs between arenas, so the run builds up power and resets afterwards.
  - Elites with telegraphed attacks come from the existing ARPG.
  - The bosses are the Archivist and the Crystal Wyrm.
- **Length:** a run takes 20 to 30 minutes.
- **Rewards and purpose:** runs unlock spells and cosmetics. That makes this the natural home for progression and for players without a full lobby.
- **Size:** L. Build it once the PvP loop and bots are solid.

10 Progression and cosmetics

No permanent power in PvP. Power lives inside a match, in the cards, and resets when the match ends. Everything earned outside a match is a new option or a new look. Party games are played by friend groups of mixed skill, and a veteran with upgraded spells makes newcomers quit.

Track	How it works
Spell unlocks	Start with six spells, and unlock the rest within the first few hours. Each unlock is a sideways option, not a stronger one.
Spell mastery	Each spell levels up with use and unlocks skins for it: colour, trail and impact.
Account level	Cosmetics and titles.
Honors	Already built in the hub: Skydiver, Stargazer and Faithful become titles and achievements.
Caldera Trials	Runs unlock spells and cosmetics (§9).

Cosmetic	Reads at the game camera	Notes
Spell skins	very well	Probably the most valued cosmetic in a spell game. Colour, trail and impact.
Headwear	very well	Silhouette. Hood, pointed hat and bare head exist today.
Robe colour and pattern	very well	Pairs with the player colour (§3).
Emotes and victory poses	very well	Party games need them, and they work with the hero's new animation set.
Dodge trail, ring-out effect	well	Short, visible, and fun to show off.
Wand or staff, name plate	somewhat	Cheap extras.

Business model: buy once, and earn cosmetics by playing. If anything is ever sold, it is cosmetic only.

11 Steam and achievements

- **Lobbies and invites.** Joining a friend takes one click from the friends list.
- **Steam's relay servers.** They remove port forwarding and hide players' IP addresses. The game already uses GameNetworkingSockets, which shares its API with Steamworks.
- **Rich presence,** for example “In the Caldera, round 3 of 5”.
- **Remote Play Together.** It needs local multiplayer, plus a shared camera when several players use one screen.
- **Controller and Steam Deck,** which is why §5 keeps the click-to-select fallback.
- **Release path:** a Steam Playtest, then a Next Fest demo with bots.

Achievement	Condition	Achievement	Condition
First Fling	Knock a rival into the lava.	**Hat Trick**	Three ring-outs in one round.
Bowling Night	Fling three players with one spell.	**Return to Sender**	Knock someone out with their own reflected spell.
Hooked	Pull a rival into the lava with Chain Hook.	**Swap Meet**	Swap a rival into the lava you were standing in.
Pacifist	Win a round without casting a spell.	**By a Thread**	Win a round with less than 10 health.

Quack Attack	Knock out a rival while they're a duck.	**Comeback Mage**	Win a match after losing the first two rounds.
Deck Builder	Stack two upgrades on one spell.	**Scholar**	Win a round with every spell.
Skydiver	Fall off Emberhold 25 times. Already an honor.	**Stargazer**	Light every constellation. Already an honor.
Faithful	Reach the cat's islet. Already an honor.	**Trial by Fire**	Finish a Caldera Trials run.

12 Roadmap

1 · M
Card pick and hotbar loadout
The new between-round screen (§4, §5).

2 · M
Colour pass
Player colours, Caldera counter-colour, brighter hub (§3).

3 · L
Bots and balance runs
Fill lobbies; overnight per-spell reports (§6).

4 · L
Spells, wave 1
Chain Hook, Firebomb, Splitting Bolt, Storm Strike, plus upgrade cards (§7).

5 · M
Frozen Lake
The first new mechanic, and map rotation (§8).

6 · M
Steam
Lobbies, invites, relay servers, rich presence, achievements (§11).

7 · M
Cosmetics and unlocks
Spell skins, robes, emotes, mastery (§10).

8 · L
Modes, then Trials
Teams, Cursed Bomb, Spellball, then the co-op roguelike (§9).

#	Phase	Done when
1	**Card pick and hotbar loadout**	In playtests, players pick a card every round without being told, and first-time players set a loadout in under 30 seconds.
2	**Colour pass**	The histogram targets in §3 are met, and a newcomer finds every player in a screenshot within a second.
3	**Bots and balance runs**	Overnight runs produce the per-spell report, and a match with one human and five bots feels like a real match.
4	**Spells, wave 1**	Every spell's relative win rate is within 0.9 to 1.1, and each new spell shows up in clips.
5	**Frozen Lake**	Playtesters describe it by its mechanic (“the ice one”), not by its colour.
6	**Steam**	A friend joins in one click without port forwarding, and achievements unlock.
7	**Cosmetics and unlocks**	Players have something to chase in their first ten hours, and no unlock changes balance.
8	**Modes, then Trials**	A solo player has a reason to start the game.

Implementation notes

Change	Where it starts in the code today
Card pick	`BoonDef`, `Boons` and `BoonStacks` in `Brawl.h` become card definitions with a type, rarity and spell. `MatchDirector::OfferBoons` filters by loadout and standing, and effects are read the way `MatchDirector::GetKnockbackScale` reads Heavy Hand today. `BrawlPhase::Shop` becomes the card phase. Protocol bump (23 today).
Hotbar drag	`SessionState::DrawSpellPicker` in `BrawlUi.cpp`, with the HUD hotbar as the drop target.
Player colours	Character tint in `DrawCharacterModel`, a rim term in `Model.frag`, and a per-player tint in the spell visuals.

Map palettes	Per-map lighting presets and a colour-grading LUT (`.cube`). Check the result with the Figure 1 histogram.
Mutators	`MatchRules` in `Pvp.h`, which already travels with the session.
Achievements	Next to the honors in `GameClient::TrackFeats`.
Bots	A server-side `BotBrain` writing `PlayerInput`, as planned in the 13 October roadmap.

Screenshots: Release build, 18 October 2026, 1600 × 900, three bots and one viewer on the Caldera, and the Emberhold hub. Colour statistics: pixels with saturation ≥ 0.2 and brightness ≥ 0.12, HUD cropped out, 24 hue bins of 15°. Icons: game-icons.net by Lorc, Delapouite and others, CC BY 3.0. Figure 3 is a mockup, not a screenshot.